# Communication

## A Deep Learning Framework for 2-D, Multifrequency Propagation Factor Estimation

Sarah E. Wessinger, Leslie N. Smith, Jacob Gull, Jonathan Gehman, Zachary Beever, and Andrew J. Kammerer

***Abstract*—Accurately estimating propagation factor over multiple frequencies within the marine atmospheric boundary layer is crucial for the effective deployment of radar technologies. Traditional parabolic equation simulations, while effective, can be computationally expensive and time-intensive, limiting their practical application. This communication explores a novel approach using deep neural networks (DNNs) to estimate the pattern propagation factor, a critical parameter for characterizing environmental impacts on signal propagation. Image-to-image translation generators designed to ingest modified refractivity data and generate predictions of pattern propagation factors over the same domain are developed. Findings demonstrate that DNNs can be trained to analyze multiple frequencies and reasonably predict the pattern propagation factor, offering an alternative to traditional methods.**

***Index Terms*—Deep neural networks (DNNs), image-to-image translation, modified refractivity, propagation factor, propagation loss.**

### I. Introduction

#### *A. Refractive Environment*

The performance of radio frequency (RF) technologies is highly susceptible to environmental factors that influence signal propagation. Fluctuations in atmospheric conditions, particularly humidity, can significantly impact signal range and quality. This variability poses challenges across diverse applications, including communication and radar technologies, which are critical to national defense as well as everyday RF devices such as cell phones and data-relay networks. Therefore, understanding and mitigating these environmental effects are crucial for optimizing the reliability and performance of RF systems.

The power at the end of an RF path is given by the one-way radio transmission equation [1], [2] as follows:

$$P_r = P_t G_t G_r \left(\frac{F}{2k_0 R}\right)^2 \tag{1}$$

where $P_r$ is the power at a distance (slant range) $R$, $P_t$ is the transmitted power, $G_t$ is the transmitter peak power gain, $G_r$ is the receiver peak power gain, $k_0$ is the RF wavenumber, and $F$ is the one-way pattern propagation factor. The only variable in (1) not determined by the radar system setup is $F$ as it encapsulates all environmental impacts on signal propagation. More specifically, $F$ is a measure of gain due to the environment. For practical applications, $F$ can also be expressed in decibels ($F_{dB}$). It is sometimes convenient to work with attenuation instead of gain using propagation loss in decibels ($P_L$) as follows:

$$P_L = 20\log(2k_0 R) - 20\log|F| \tag{2}$$

$$F_{\mathrm{dB}} = 10\log|F|. \tag{3}$$

Accurately estimating $F$ (or, equivalently, $P_L$) is imperative for quantifying environmental attenuation on RF signals. Current methods of estimating $F$ rely on parabolic equation simulations, such as the advanced propagation model (APM; [3]), the variable terrain radio parabolic equation (VTRPE; [4]), and the tropospheric electromagnetic parabolic equation routine (TEMPER; [5]). These simulations require inputs of modified refractivity ($M$), which is calculated from atmospheric refractivity using temperature in Kelvin ($T$), partial vapor pressure in millibar ($e$), and pressure in millibar ($p$), and adjusted to account for Earth's curvature [6] as follows:

$$M(z) = \left(\frac{77.6\,p}{T} + \frac{373256\,e}{T^2}\right) + \frac{z}{R_e} \times 10^6 \tag{4}$$

where $z$ is altitude and $R_e$ is Earth's radius. This study explores using deep neural networks (DNNs) to predict $F$ based on $M$. DNNs excel at approximating complex nonlinear relationships by performing regressions on extensive datasets. While computationally intensive to train, once trained, DNNs offer significant advantages in terms of speed and efficiency.

DNNs and other deep learning techniques have been previously applied to RF technologies, particularly for communication applications [7], [8], [9], [10], [11], [12]. Notably, Bal et al. [10] demonstrated the potential of DNNs in estimating the path loss exponent for wireless communication; Bakirtzis et al. [11] used CNNs to estimate the received signal strength of communication devices from images of physical indoor environments and demonstrated quicker computation times compared to traditional methods; and Huang et al. [12] used a CNN to solve the parabolic wave equations to estimate received signal strength over irregular terrain. However, to the authors' knowledge, no prior studies have applied DNNs to estimate $F$, $P_L$, or $F_{dB}$ over the entire domain of range and altitude.

DNNs have been used within radar applications for both forward and inverse problems [13], [14], [15], [16], [17]. Forward problems estimate the refractive environment using environmental data, while inverse problems estimate environmental conditions or perform object detection using the refractive environment. Estimating evaporation duct height (EDH) is a popular multivariate forward problem within the refractive environment, with prior studies employing DNNs [14], [16], [17]. Evaporation ducts, caused by steep near-surface vertical humidity gradients, are of interest because they trap RF waves and extend signal ranges at altitudes below the EDH. Evaporation ducts form within ~30 m of the ocean surface; this near-surface region is often turbulent, resulting in a high degree of self-similarity among evaporation ducts within $M(z)$ profiles. Thus, estimating EDH is a generally easier problem for DNNs than learning RF signal propagation behavior based on any $M(z)$. This study focuses on a

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forward problem where numerical weather prediction data is used to estimate the refractive environment through $F$ or $P_L$.

Characterizing RF signal propagation is especially challenging in the marine atmospheric boundary layer due to constantly fluctuating thermodynamic variables. RF technologies operating in this environment commonly use S- and X-band frequencies; thus, accurately estimating $F$ across both bands is essential. Prior studies using DNNs for multifrequency RF problems commonly train separate models for each frequency [17], [18]. In contrast, this study trains a single DNN on a dataset comprised of two frequencies and then compares the two-frequency predictions to the individual frequency predictions. Additionally, prior research has primarily focused on attenuation as a function of only range or altitude but not both simultaneously [17], [19], [20], [21]. To the authors' knowledge, estimating $F$ simultaneously over both range and altitude using DNNs has not yet been attempted.

This study makes three novel contributions: 1) it demonstrates the capability of a trained DNN to accurately estimate the pattern propagation factor; 2) it achieves this estimation over both altitude and range; and 3) it introduces a unified DNN framework capable of accounting for multiple frequencies.

### *B. Image-to-Image Translation*

Image-to-image translation is a powerful machine learning technique that enables the transformation of an image from one domain, characterized by a specific set of attributes, to another domain with new attributes, while preserving the essential content of the original image [22], [23]. This process involves learning the intricate mapping between the two domains, allowing for a wide range of applications in image processing, including image generation, style transfer, and segmentation [24]. In this context, the $M$ domain (i.e., modified refractivity) is the input "image" and the $F$ domain is the desired output "image."

In recent years, deep learning frameworks have revolutionized the field of image-to-image translation, with autoencoders, convolutional neural networks (CNNs), and generative adversarial networks (GANs) achieving remarkable success. GANs, in particular, have emerged as a popular choice for image-to-image translation tasks [24], [25], due to their ability to learn complex mappings between domains. A typical GAN architecture for image-to-image translation consists of two primary components: a generator network and a discriminator network. The generator network produces an image in the output domain, while the discriminator network evaluates the generated image, providing feedback to the generator to improve its performance.

### *C. Neural Network-Based Emulator Architecture*

Initially, GAN architecture was implemented for image-to-image translation, as it is a widely accepted architecture for this task. The discriminator in a GAN plays a crucial role in guiding the generator by assessing the realism of the generated images. Through adversarial training, the discriminator enforces realism by distinguishing fine details, textures, and patterns that align with the desired target distribution. This feedback loop enables the generator to iteratively refine its outputs, ensuring they match the statistical properties of the real data. However, in this study, supervised training with explicitly desired outputs renders the adversarial training unnecessary and only the generator is needed.

U-Net architectures [26] are commonly employed as the generator in GANs [26]. Their architecture is characterized by a contracting and an expanding block, connected by skip connections and an inner module. Skip connections enable the model to detect features at multiple resolutions, capturing both global structural relationships and pixel-wise details. This design allows the U-Net to effectively learn the complex mappings between domains, making it an ideal choice for image-to-image translation tasks.

This study evaluated the merits of implementing a variety of U-Net-based generator architectures. The following lists the models tested in order of increasing size and complexity.

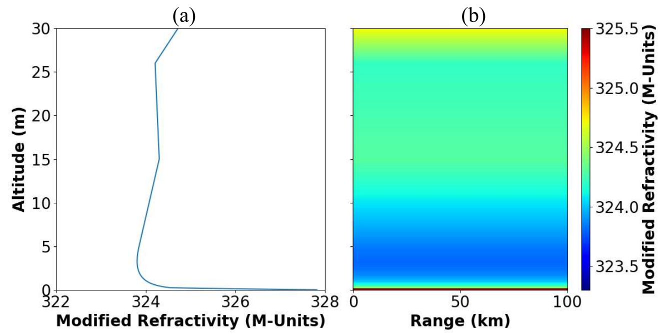


Fig. 1. (a) $M(z)$ used to create the (b) range-homogeneous $M$ domain input for RefractNet.

1) Two versions of U-Net [26].
2) Recursive neural network (RNN).
3) Efficient U-Net [27].
4) Dilated U-Net [28].
5) UVCGANv1 and v2 (visual transformer at bottleneck) [29], [30].
6) UCTransNet (transformer instead of skip connections) [31].
7) A novel hybrid (combines UVCGan and UCTransNet).

We observe significant improvements in performance from the first U-Net's test to the larger and more complex models. However, our evaluations demonstrate that architectural innovations alone are approaching the limits of significant performance gains. While initial gains underscore the importance of exploring alternative architectures, the final two models only provide minor performance improvements but with an increase in computational cost. Consequently, UVCGANv2 is the default model used in this study. The UVCGANv2 architecture, henceforth referred to as "RefractNet," is illustrated in [30]. RefractNet's training hyperparameters are listed in Table I for the sake of reproducibility. The Bayesian optimization method Optuna is used to tune these parameters.

## II. Methods

### *A. Data*

$M(z)$ used within this study is from the Naval Research Laboratory's numerical weather prediction system, the Coupled Ocean Atmosphere Mesoscale Prediction System (COAMPS). COAMPS forecasts are run at seven arbitrary oceanic locations across the globe to remove local geographic dependence on these results.

The physics-based parabolic simulation used to calculate the refractive domain of $F$ from $M(z)$ is TEMPER version 3.2.3 for two frequencies, 3 and 10 GHz (S- and X-Band, respectively). TEMPER's mesh size to calculate F is approximately half the frequency wavelength, assuming a smooth sea surface. Rough ocean surfaces could be an added complexity at a later time; the point of this study is an initial proof-of-concept focusing on refractivity, $M(z)$. The antenna height is set to 20 m for each simulation.

TEMPER simulations for S- and X-band have varying domain output altitudes based on mean sea level. All S-band runs range in altitude from mean sea level (0 m) to 300 m, whereas X-band runs extend from mean sea level (0 m) to 30 m, both at a 0.2-m vertical resolution. All S-band runs have a horizontal range from 0 to 210 km at a 0.2-km resolution, whereas X-band runs extend from 0 to 100 km, at a 0.1-km resolution.

This study used 70 322 total cases, comprised of an $M(z)$ input and an $F$ domain output to train RefractNet, and 15 070 total cases are used to evaluate its performance. Half of all cases, for both test and train datasets, are X-band and half are S-band. We expect this number of training samples to be sufficient to train our network because several well-known image benchmark datasets contain on the order of 50 000 training samples (i.e., MNIST and Cifar-10/100).

TABLE I
REFRACTNET TRAINING HYPERPARAMETERS

| Hyper-parameter | Value |
|---|---|
| Batch size | 140 |
| Num Epochs | 100 or 200 |
| Learning rate (LR) | 0.0001 |
| LR schedule | 1cycle |
| Dropout | 0.02 |
| Num uvcgan2 blocks | 12 |
| Num uvcgan features | 384 |

Note, $M$ is assumed homogeneous over range and time for this study (Fig. 1).

## B. Metrics

In addition to visual inspection of individual examples, this study leveraged a combination of three key performance metrics: mean squared error (mse), structural similarity (SSIM), and Fréchet inception distance (FID). MSE is a widely used metric that calculates the average squared difference between the pixel values of real and generated images, penalizing larger errors more heavily. However, as a pixel-wise measure, mse is limited in capturing larger-scale structural trends, which are critical in evaluating the accuracy of complex images. In contrast, SSIM [32] assesses error while considering structural information and broader trends, yielding a score between 0 (no similarity) and 1 (perfect similarity). Utilizing convolutional properties, SSIM effectively identifies low-level structures within images, which is critical for evaluating the accuracy of the main structure. FID [33] is the most common metric used to evaluate GAN results as it compares features extracted from real and generated images using a pre-trained network, with lower scores indicating better fidelity to real images. SSIM and FID are particularly useful in evaluating the performance of the emulator as they provide a more comprehensive and nuanced understanding of the output quality.

Each metric can be computed for individual samples, batches, frequency, and depth categories, enabling clustering analyses to identify classes of inputs where the model performs poorly. This is particularly useful in identifying areas where the model requires improvement and guides the approach used to optimize the emulator's performance. As such, during RefractNet's development, these metrics-guided efforts improve the model's accuracy through various methods, including data augmentation, loss function optimization, hyperparameter tuning, and experimentation with different U-Net architectures. We tested several loss function variations and settled on an equal combination of MSE and SSIM for our loss function.

# III. EXPERIMENTS

## A. Setup

Eight experiments are conducted during this study (Table II). Input/output domains having 256 × 256 pixels are optimal for a balance between performance and speed. To meet this requirement, $M(z)$ is interpolated to have 256 equally spaced points over the evaluated altitudes (i.e., 0–30 m or 0–300 m) and repeated 256× over range (Fig. 1). The corresponding domains of $F$ are interpolated to have the same points over altitude as the $M$ domains (256 × 256) and equally spaced points over range between 0 and 100 km, so both frequencies are evaluated over the same physical space. Input and output domains are scaled to have values between 0 and 1 due to batch normalization layers within the architecture. As such, all data points for $M$ domain images and $F$ domain images are normalized based on altitude and gain variable following the normalization schemes in Table III based on (5), where $V$ is the variable being normalized ($M$, $F$, etc.) as follows:

TABLE II
COMBINATIONS OF VARIABLES USED TO TRAIN THE REFRACTNET FOR EACH EXPERIMENT

| Experiment Number | Variable, Altitude, Frequencies |
|---|---|
| Experiment 1 | $F$, 30 m, X + S |
| Experiment 2 | $F$, 30 m, X |
| Experiment 3 | $F$, 30 m, S |
| Experiment 4 | $F_{dB}$, 30 m, X + S |
| Experiment 5 | $F_{dB}$, 30 m, X |
| Experiment 6 | $F_{dB}$, 30 m, S |
| Experiment 7 | $F$, 300 m, S |
| Experiment 8 | $F_{dB}$, 300 m, S |

TABLE III
NORMALIZATION SCHEMES USED BASED ON ALTITUDE AND GAIN VARIABLE

| Altitude, Variable | $M(z)$ Norm. | $F(z,x)$ Norm. |
|---|---|---|
| 30 m, $F$ | $M = \frac{M-288}{181}$ | $F = \frac{F}{16.45}$ |
| 300 m, $F$ | $M = \frac{M-282}{187}$ | $F = \frac{F}{16.45}$ |
| 30 m, $F_{dB}$ | $M = \frac{M-288}{181}$ | $F_{dB} = \frac{F_{dB}+90.01}{-102.17}$ |
| 300 m, $F_{dB}$ | $M = \frac{M-282}{187}$ | $F_{dB} = \frac{F_{dB}+90.01}{-102.17}$ |

$$V(z,x) = \frac{V(z,x) - V_{\min}}{V_{\max} - V_{\min}}. \tag{5}$$

In addition, single-frequency experiments have half the number of training samples as the dual-frequency experiments, as previously described. Thus, RefractNet is trained for twice as many epochs for single-frequency experiments in order to minimize any training advantage the dual-frequency experiments might gain by having twice as many training samples.

## B. Results

Each evaluation metric (MSE, FID, and SSIM) is calculated between the RefractNet-predicted $F$ domain image and TEMPER's normalized $F$ domain image for each test case in each experiment (Fig. 2), thus these values are dimensionless. Distributions of evaluation metrics, presented in Fig. 3, demonstrate that RefractNet performs well across a variety of scenarios. The majority of cases exhibit high SSIM values (indicating strong SSIM between predictions and TEMPER) and low FID and MSE values (demonstrating low dissimilarity and error). Fig. 4 illustrates a case that performed poorly due to having ducting (trapping) features at longer ranges, which are indicative of signal interference due to environmental reflection and/or refraction.

To rigorously assess RefractNet's capacity to handle multiple frequencies, performance of single-frequency experiments against their dual-frequency counterparts is statistically compared. Specifically, two-sample t-tests are performed on the MSE, SSIM, and FID populations for the following experiment pairs, ensuring consistent altitude and gain variables: Experiments 1 and 2, Experiments 1 and 3, Experiments 4 and 5, and Experiments 4 and 6. P-values from Experiments 1 and 2, Experiments 1 and 3, and Experiments 4 and 5 (FID and MSE only) of the t-tests are $< 0.05$, indicating these null hypotheses (e.g., average MSE Experiment 1 = average MSE Experiment 3) are rejected, and the average values (Table IV) of these evaluation metrics are different in a statistically significant sense. Interestingly, the null hypotheses in the case of Experiments 4 and 6 are accepted, suggesting these average values are not statistically significantly different.

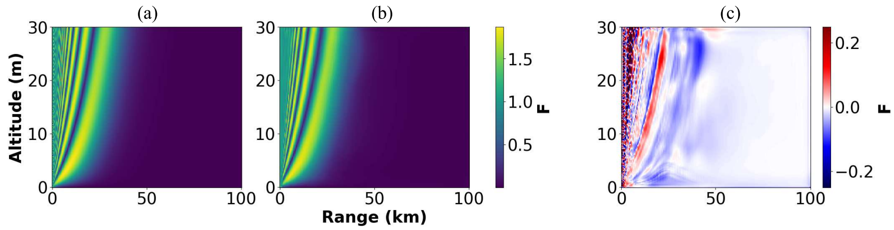


Fig. 2. $F$ domain calculated by (a) TEMPER, (b) RefractNet, and (c) the difference between these domains, from a case in Experiment 1 where the evaluation metrics are MSE = $1.086 \times 10^{-5}$, FID = 0.0016, and SSIM = 0.930.

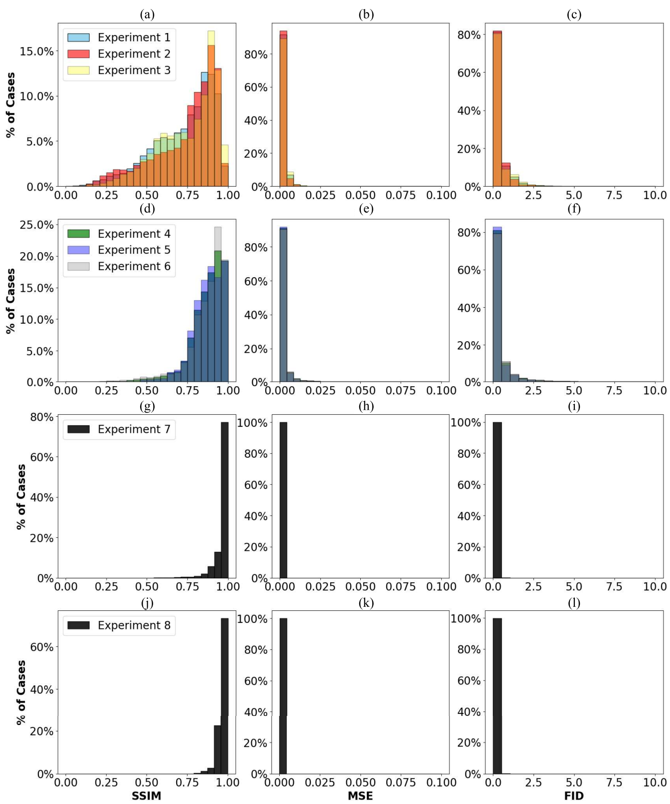


Fig. 3. Distribution of evaluation metrics for SSIM (a), (d), (g), and (j); MSE (b), (e), (h), and (k); and FID (c), (f), (i), and (l) for Experiments 1–3 (a)–(c), 4–6 (d)–(f), 7 (g)–(i), and 8 (j)–(l).

TABLE IV

AVERAGE VALUES OF EVALUATION METRICS FOR EACH EXPERIMENT

| Experiment Number | MSE | FID | SSIM |
|---|---|---|---|
| Experiment 1 | 0.0010 | 0.259 | 0.724 |
| Experiment 2 | 0.0009 | 0.230 | 0.742 |
| Experiment 3 | 0.0011 | 0.284 | 0.752 |
| Experiment 4 | 0.0017 | 0.439 | 0.871 |
| Experiment 5 | 0.0016 | 0.397 | 0.873 |
| Experiment 6 | 0.0017 | 0.442 | 0.870 |
| Experiment 7 | 0.0001 | 0.010 | 0.965 |
| Experiment 8 | 0.0001 | 0.025 | 0.971 |

X-band-only experiments (Experiments 2 and 5) consistently outperformed both S-band-only (Experiments 3 and 6) and dual-frequency experiments (Experiments 1 and 4) across all metrics (Table IV), albeit not by much. This highlights an important consideration: RF signal propagation becomes increasingly sensitive to atmospheric refractive structures at higher frequencies [34]. Consequently, the $F$ domain exhibits more distinct and repetitive patterns at higher frequencies, potentially contributing to more robust and effective DNN training.

TABLE V

AVERAGE VALUES OF EVALUATION METRICS FOR EACH FREQUENCY IN DUAL FREQUENCY EXPERIMENTS

| Experiment Number, Frequency | MSE | FID | SSIM |
|---|---|---|---|
| Experiment 1, X | 0.0010 | 0.235 | 0.706 |
| Experiment 1, S | 0.0011 | 0.283 | 0.743 |
| Experiment 4, X | 0.0016 | 0.408 | 0.873 |
| Experiment 4, S | 0.0018 | 0.469 | 0.869 |

Despite evaluation metrics performing better for X-band-only training, dual-frequency-trained experiments yield similar patterns within the $F$ and $F_{dB}$ domains (Fig. 5) with similar average evaluation metric values for each individual frequency in the dual-frequency experiments (Table V) compared to the single-frequency-trained experiments. Thus, this DNN framework is able to account for two frequencies.

To further investigate RefractNet's performance, predictions of $F$ versus $F_{dB}$ domains are evaluated. RefractNet generally performed better when predicting $F$ as shown by average MSEs and FIDs. This difference likely arises from the logarithmic conversion from $F$ to $F_{dB}$ [(3)], which amplifies the influence of refractive structures present in the refractive domains (as evidenced by the larger average SSIM values), especially at longer ranges (Fig. 4), during training. This suggests that DNNs should be trained to estimate $F$, with subsequent calculation of desired gain or attenuation variables (e.g., $F_{dB}$ and $P_L$) derived from the predicted $F$ values.

To explore this suggestion, RefractNet's predicted $F$ domain images from Experiment 1 (i.e., trained on $F$) are converted to $F_{dB}$ [(3)], and compared to TEMPER's $F_{dB}$ domain images using the evaluation metrics. These evaluation metrics are compared to Experiment 4 (Fig. 6), which is trained on $F_{dB}$ and compared RefractNet's predicted $F_{dB}$ domain images to TEMPER's $F_{dB}$ images. Results show images of $F_{dB}$ converted from RefractNet-generated $F$ generally have greater SSIM to TEMPER's predicted $F_{dB}$ domain images than Experiment 4, but greater errors in estimated values (Fig. 7).

To explore RefractNet's performance over various altitudes, all S-band-only experiments are compared. Higher-altitude experiments (Experiments 7 and 8) consistently outperformed lower-altitude ones (Experiments 3 and 6). This finding aligns with expectations, as higher-altitude scenarios contain more data points within the radar horizon and fully resolve patterns of constructive interference that occur over 30 m (Fig. 8). Conversely, lower-altitude scenarios contain more pixels outside of the radar horizon and do not fully resolve propagation structures occurring over 30 m (Fig. 4). Therefore, future studies developing DNNs for radar applications, which are aimed at identifying specific propagation features, should carefully consider over what altitudes and ranges to evaluate and potentially even exclude data points within the radar horizon to enhance model training and accuracy.

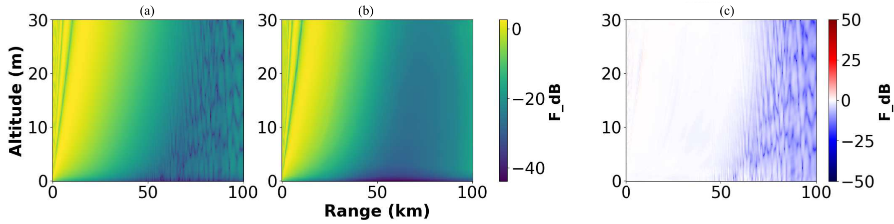


Fig. 4. $F_{dB}$ domain calculated by (a) TEMPER, (b) RefractNet, and (c) the difference between these two domains for a case from Experiment 6 where the evaluation metrics are MSE = 0.0020, FID = 0.515, and SSIM = 0.758.

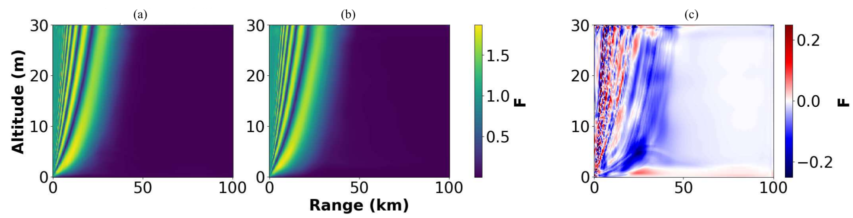


Fig. 5. $F$ domain for the same case in both (a) Experiment 1 with metrics of MSE = $1.086 \times 10^{-5}$, FID = 0.0016, and SSIM = 0.930; (b) Experiment 2 of MSE = $8.035 \times 10^{-6}$, FID = 0.0012, and SSIM = 0.843; and (c) the difference between these domains.

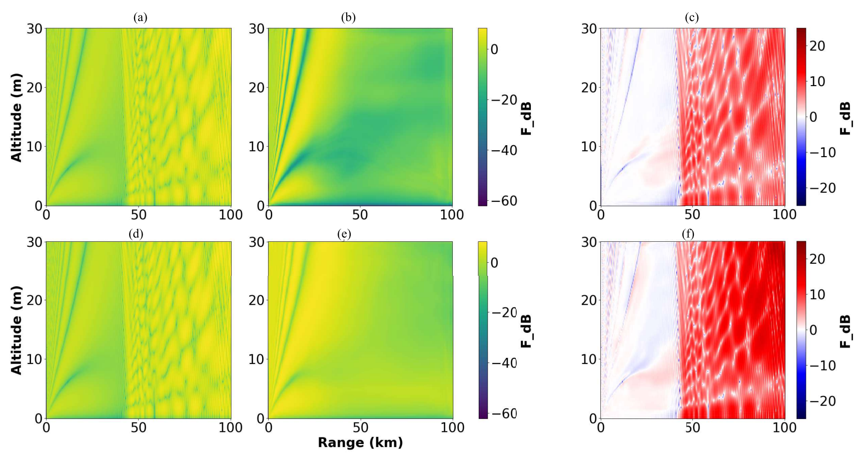


Fig. 6. $F_{dB}$ domain (a) and (d) from TEMPER, (b) calculated from Experiment 1 $F$, (c) the difference between a and b, (e) from Experiment 4, and (f) the difference between d and e.

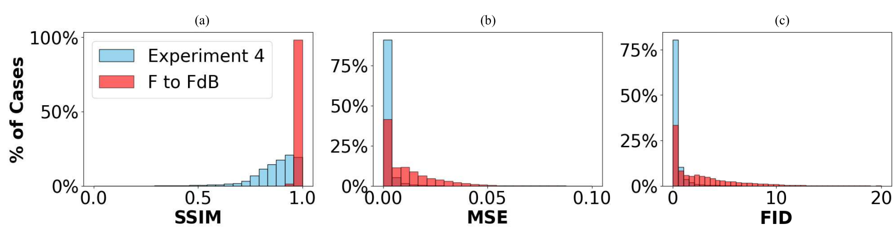


Fig. 7. Distributions of (a) SSIM, (b) MSE, and (c) FID for Experiment 4 compared to the calculated $F_{dB}$ from Experiment 1 $F$.

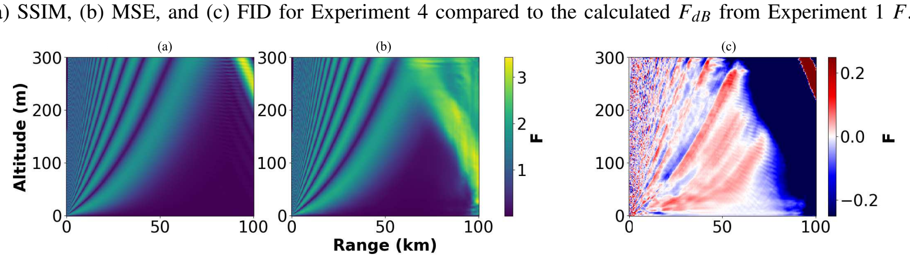


Fig. 8. $F$ domain calculated by (a) TEMPER, (b) RefractNet, and (c) the difference between these domains for a case from Experiment 7 where the evaluation metrics are MSE = 0.0021, FID = 0.505, and SSIM = 0.749.

## IV. Conclusion

This study introduced RefractNet, an image-to-image DNN generator designed for efficient simulation of the propagation factor, over both range and altitude, while accounting for multiple frequencies. RefractNet exhibited a slightly superior performance for X-band frequencies compared to S-band. From a practical standpoint, RefractNet demonstrated the capacity to handle both frequencies simultaneously. These results not only hold significant promise for the future of trained DNNs as a viable method to estimate propagation factor, but also illustrate the sensitivity of predicting propagation factor using DNNs. Results of this study varied considerably between training on different domain altitudes, $F$ or $F_{dB}$, single or multiple frequencies, and even training on $F$ and assessing performance using $F_{dB}$. Thus, future work estimating the propagation factor using DNNs needs to carefully consider how training is approached.

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